

Assignment 2 SECPH 2523

QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offences by students in a college.

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
				00-690	15/03/06	1	20.00
				00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
				00-770	19/06/06	2	30.00

a) Table 2.1 is said to be unnormalized table. Why?

b) Transform the unnormalized Table 2.1 into a First Normal Form (1NF) table. Show the transformed 1 NF table. What is the primary key for this new 1 NF table.

a) Some cells have multiple value.

b) 1 NF

StudentID	Name	LicenseNo	CarReg	ReCeiptNo	Date	OffenceCode	Fine (RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
F1011	CK Fong	P100	JGK9201	00-690	15/03/06	1	20.00
F1011	CK Fong	P100	JGK9201	00-810	20/03/06	2	30.00
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QUESTION 2

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
		12	Pauline James	B	\$50
		16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
		17	Monique Williams	B	\$50

1NF

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
1023	Madagascar travel site	12	Pauline James	B	\$50
1023	Madagascar travel site	16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
1056	Online estate agency	17	Monique Williams	B	\$50

2NF

Project number (PK)	Project name
1023	Madagascar travel site
1056	Online estate agency

Employee number (PK)	Employee name	Rate category	Hourly rate
11	Vincent Radebe	A	\$60
12	Pauline James	B	\$50
16	Charles Ramoraz	C	\$40
17	Monique Williams	B	\$50

3NF – reduce redundancy but still connecting

Project number (PK)	Project name
1023	Madagascar travel site
1056	Online estate agency

Employee number (PK)	Employee name
11	Vincent Radebe
12	Pauline James
16	Charles Ramoraz
17	Monique Williams

Rate category (PK)	Hourly rate
A	\$60
B	\$50
C	\$40

Project number	Employee number	Rate category
1023	11	A
1023	12	B
1023	16	C
1056	11	A
1056	17	B